

Teacher-Anchored Selection of Post-Training Quantized Models under Domain Shift

Alejandro Rodriguez Dominguez,^{1,2,3} Muhammad Shahzad,¹ Xia Hong¹

¹Department of Computer Science, University of Reading, Reading, United Kingdom

²Quantitative Analysis and Artificial Intelligence Department, Miralta Finance Bank S.A., Madrid, Spain

³Lecturer, Albert School, Madrid, Spain

Abstract

Compressing a trained model yields a family of deployment candidates, and under domain shift the most compressed one need not be the one to deploy. We study selection over such a family, with candidates and teacher fixed and target labels absent or scarce. Two findings organize the label-free case. Minimum teacher distortion behaves almost as a constant rule, selecting the same eight-bit, per-channel, unclipped configuration in every run, which does not minimize empirical target cross-entropy. Established estimators divide sharply: in the overconfident-collapse regime of the CNN families, confidence-based estimators order the family close to backwards, and the diagnostics that identify it need the labels the setting denies, while output-distribution estimators match the teacher-relative anchor and on one architecture beat it. Distortion is nonetheless stable, so a supervised term can move selection away from it. Combining the two, we give exact quadratic identities for a canonical quadratic analogue of the family. We also show that under symmetric corruption the label-dependent part of a criterion linear in the label indicator is multiplied by one common factor whenever its coefficient sums are candidate-invariant, a class holding teacher contrasts and accuracy but not cross-entropy. These characterize the score’s components without bounding selection regret. Across one hundred and thirty-four candidate families, one per independently trained convolutional or Vision Transformer teacher, anchoring reduces mean regret at the smallest label budget in every setting, an advantage that fades beyond twenty-five labels.

1 Introduction

Model compression usually produces several plausible deployment candidates rather than one preferable model (Cheng et al. 2018). Post-training quantization is a clear example (Jacob et al. 2018), where bit widths, clipping thresholds, granularity and layer configurations generate a family derived from one full-precision teacher (Gou et al. 2021; Hinton, Vinyals, and Dean 2015), and recent methods keep enlarging it by tuning parameters per layer (Ranjan and Savakis 2026; Arai and Ichikawa 2025). Under domain shift, candidates that behave alike on source or calibration data can have very different target risks (Ovadia et al. 2019; Minderer et al. 2021), so the construction criterion need not identify the best model to deploy.

This is a deployment-time problem distinct from candidate construction: given an already built family sharing a teacher,

which candidate should be used on the target domain? Candidates are neither retrained nor adapted during selection, and the held-out target test set is used only for evaluation. The objective is the candidate with lowest target cross-entropy. Operational constraints such as memory, latency or energy are assumed to have restricted the family beforehand, so we study selection within an admissible family, not the broader problem of trading risk against cost.

Two signals are available and neither suffices alone. Distance to the shared teacher needs no labels and is stable across resamples (Liu et al. 2023), but on standard grids the candidate closest to its teacher is almost always the least-compressed one, so its ranking is nearly fixed in advance. A small labeled target sample measures the objective directly but at the budgets a practitioner will spend is dominated by variance (Okanovic et al. 2025), and corrupted labels degrade its ordering further (Chen et al. 2021). The failures are complementary, one biased but stable and the other unbiased but noisy, which is where shrinkage should help.

Anchored selection makes that shrinkage explicit: the label-free distortion is the base score, the supervised loss enters as a weighted correction, and the weight is chosen by cross-validation inside the labeled sample, so the labels move the choice away from the teacher only as far as they support. The construction admits exact analysis. Quadratic identities relate the members of a canonical quadratic analogue of the family and make explicit what each supervised term adds beyond the teacher distance; the selector we evaluate replaces the squared distortion with KL and tunes its coefficient, so the identities characterize components without governing regret. Under symmetric corruption the label-dependent part of a criterion linear in the label indicator attenuates by a single common factor whenever its coefficient sums do not vary across candidates. That covers validation accuracy as well as the Brier score (Gneiting and Raftery 2007), so the result describes a class rather than favouring any member of it.

Our contributions are to formulate deployment-time selection over a fixed family sharing a teacher and to characterize its label-free baselines, showing that the teacher-relative rule is close to degenerate, that confidence-based estimators can fail catastrophically under overconfident quantization collapse in a way that depends on a regime not identifiable without target labels, and that output-distribution estimators remain competitive with the anchor and on one architec-

ture outperform it; to develop the anchored family, together with quadratic identities for its canonical quadratic analogue; to prove the attenuation result; and to show that anchoring improves on direct validation at the smallest label budgets, an advantage that fades as the supervised estimate becomes reliable alone.

2 Related Work

Performance estimation under shift. Without target labels, methods estimate performance from confidence, entropy, calibrated thresholds, class dispersity or prediction-matrix statistics (Hendrycks and Gimpel 2017; Garg et al. 2022; Guillory et al. 2021; Deng et al. 2023; Xie et al. 2024). These target absolute performance, a weaker requirement than ours: selection needs only the ordering, but needs it right between candidates that differ slightly, so small estimation errors reverse it. A second line reads agreement between independently trained predictors as a proxy for accuracy (Jiang et al. 2022; Baek et al. 2022), and concurrent work supplies a foundation model as external reference (Li et al. 2026). Both require a reference the practitioner must provide; ours already contains one, since every candidate derives from the same teacher. Semi-Supervised Model Evaluation is closest in the supervision it assumes (Shanmugam et al. 2025), but models the joint scores of several classifiers rather than their structured relation to a common parent.

Shared teachers and compressed-model construction. The teacher those methods lack is standard in compression. Distillation transfers a teacher’s predictive distribution to a smaller model (Hinton, Vinyals, and Dean 2015; Tian, Krishnan, and Isola 2020), and post-training quantization methods use prediction differences, layer sensitivities or propagated quantization errors to choose quantization parameters (Liu et al. 2023; Ranjan and Savakis 2026; Arai and Ichikawa 2025). There the teacher guides construction and the resulting candidate is the answer; our problem begins with the family fixed. Teacher distance is the signal to carry across that boundary, and carrying it alone is not enough.

Selection with scarce or unreliable labels. Spending labels raises the question of what to compute from very few. Direct validation is the natural baseline, and label-efficient selection has been approached by choosing which examples to annotate (Okanovic et al. 2025); ours is complementary, since every selector receives the same examples and budget. Which loss is computed matters, because cross-entropy, Brier score and accuracy induce different rankings once calibration and confidence shift (Guo et al. 2017; Ovadia et al. 2019; Minderer et al. 2021). The Brier and logarithmic scores are standard proper scoring rules (Gneiting and Raftery 2007); we introduce no new one, asking instead which to anchor. Under multiclass class-conditional noise validation accuracy can retain selection information (Chen et al. 2021); our attenuation result explains why it survives corruption, and identifies the class of linear criteria that share that behavior. Anchored selection sits at the intersection of these three. Performance estimation normally imports a reference from outside; here construction has already supplied one,

the teacher, which we combine with a supervised term small enough that the scarce-label concerns apply directly.

3 Anchored Selection and Its Identities

Let f be a fixed teacher with predictive distribution $p_f(\cdot | x)$ and $\mathcal{G} = \{g_1, \dots, g_M\}$ a finite family of candidates with predictions $p_g(\cdot | x)$. The framework assumes only that all candidates share the same reference teacher. On the shifted target domain we observe an unlabeled pool $\mathcal{U}$ and, depending on the regime, a labeled sample $\mathcal{S} = \{(x_i, y_i)\}_{i=1}^n$. A disjoint test set $\mathcal{T}$ is used only for evaluation and defines $L_{\text{test}}(g) = |\mathcal{T}|^{-1} \sum_{(x,y) \in \mathcal{T}} -\log p_g(y | x)$. Selector quality is the cross-entropy selection regret $R(\hat{g}) = L_{\text{test}}(\hat{g}) - \min_g L_{\text{test}}(g)$.

One Family of Selectors

Without target labels the natural rule keeps the candidate whose predictions stay closest to the teacher’s, by the average KL distortion over the unlabeled pool,

$$D(g) = \frac{1}{|\mathcal{U}|} \sum_{x \in \mathcal{U}} \text{KL}(p_f(\cdot | x) \| p_g(\cdot | x))$$

which gives $\hat{g}_D = \arg \min_g D(g)$. It measures how much a candidate changes the teacher’s predictions, not whether the change is favorable.

Every selector we study adds one function of the labeled sample to that base score:

$$S(g) = D(g) + \alpha \Phi_{\mathcal{S}}(g), \quad \alpha \geq 0 \quad (1)$$

so the members differ only in the choice of $\Phi_{\mathcal{S}}$, and $\alpha = 0$ recovers pure distortion. Three choices matter here.

- **CE-combo** takes $\Phi_{\mathcal{S}} = \widehat{\text{CE}}_{\mathcal{S}}$, the empirical cross-entropy on the labeled sample. Large α recovers direct validation, so (1) interpolates between the two label regimes.
- **Alignment** takes $\Phi_{\mathcal{S}} = -A_{\mathcal{S}}$, where $A_{\mathcal{S}} = |\mathcal{S}|^{-1} \sum \langle p_g(x) - p_f(x), e_y - p_f(x) \rangle$ scores how far a candidate moves *from* the teacher prediction *toward* the observed class, rather than how far it moves at all. The sign is flipped because favorable movement should lower the score.
- **Permutation control** takes the same $A_{\mathcal{S}}$ after permuting the observed labels across the sample. It keeps the statistic’s scale and destroys only the label–input correspondence. What it isolates is therefore what the direction contributes beyond the shrinkage.

Tables abbreviate these consistently as Dist. for the label-free anchor, Val-CE and Val-Acc for direct validation on cross-entropy and on accuracy, CE-combo, Align, Perm. for the permutation control and Teach. for the teacher component. The coefficient is always chosen by cross-validation inside $\mathcal{S}$ with $\alpha = 0$ in the grid, so no selector receives labels the others do not, and each can revert to the anchor when the data do not support leaving it. Figure 1(a) shows the geometry.

What the Family Implies

Two consequences of writing the selectors this way are used later; derivations are in Supplement A.

Alignment separates into a label part and a label-free part. Because A_S is linear in the observed class, $A_S = A_S^{\text{lab}} + A_S^{\text{teach}}$, where only the first uses e_y . Retaining the second alone gives the *teacher component*. In the fully quadratic identity the corresponding teacher-only criterion reduces to predictive-concentration ranking; the implemented ablation, which keeps KL distortion and a cross-validated coefficient, is related but not identical. The selector is not label-free: its coefficient is still cross-validated on the labeled sample, so only the score component is label-independent. We evaluate it as an ablation.

Label information decays at one rate for a class of criteria. Write a label-dependent criterion as $a_g(x)^\top e_y$. Under symmetric K -class corruption that retains the clean label with probability $1 - \eta$,

$$\mathbb{E}[a_g^\top e_{\tilde{y}} | y] = \left(1 - \frac{K}{K-1}\eta\right) a_g^\top e_y + \frac{\eta}{K-1} \mathbf{1}^\top a_g,$$

so a single multiplicative factor governs candidate *comparisons* exactly when the offset $\mathbf{1}^\top a_g$ does not vary across candidates. That condition holds for teacher contrasts such as $p_g - p_f$, whose coordinates sum to zero, and for accuracy indicators, whose coordinates sum to one, giving $\mathbb{E}[\widehat{\text{Acc}}^{\text{noisy}}] = (1 - \frac{K}{K-1}\eta)\widehat{\text{Acc}} + \frac{\eta}{K-1}$. It does not hold in general: the coefficient vector of cross-entropy, $-\log p_g$, has a candidate-dependent coordinate sum. The alignment statistic and validation accuracy therefore share one attenuation rate, and the result singles out neither of them; it says nothing about criteria outside that class.

4 Empirical Evaluation

Four questions organize this section. Can a label-free rule solve the problem at all? Does spending few labels help? Does the directional term add anything the anchoring does not? And does the corruption analysis predict what we measure?

We evaluate three target-supervision regimes: no labels, scarce clean labels, and corrupted labels. Candidate parameters are never updated. Two resources are held fixed: the target inputs, which every selector may use in full, and the label budget, the same n examples for each supervised selector including those its coefficient selection consumes.

Settings and protocol. A *run* is one teacher with its fixed 72-candidate family; an *execution* repeats the pipeline with a retrained teacher on identical splits, so a shift-seed configuration recurs once per execution with a different candidate family. There are 74 base configurations, 15 and 15 on the CNNs and 17 and 27 on the ViTs; since the CNN settings were executed three times and the ViT settings once, the number of independently trained teachers, hence of candidate families, is $45 + 45 + 17 + 27 = 134$. The statistical unit throughout is the run, so pooled CNN comparisons have 45 units per architecture; Supplement F documents the runs excluded from this cohort and the structural criteria used. Four settings are used throughout; Supplement Table S4 summarizes them

and Supplement Table S2 gives the full accounting including within-run repetitions. The DomainNet (Peng et al. 2019) CNN experiments use five fixed shifts at 30 classes, from Real to Clipart, Painting and Sketch, and from Clipart and Painting to Sketch. ResNet50 (He et al. 2016) and MobileNetV2 (Sandler et al. 2018) use three seeds per shift, giving 15 base configurations each. Their teachers come from transfer learning on ImageNet (Russakovsky et al. 2015), and each architecture is run through three independent end-to-end executions on identical splits. The two Vision Transformer (Dosovitskiy et al. 2021) settings are DomainNet at 50 classes (17 runs) and held-out-domain CIFAR-20 (Krizhevsky 2009) (27 runs), with teachers from a continual-distillation pipeline; because teacher construction differs between the two groups, we do not read cross-setting differences as evidence about teacher quality alone. Selection pools contain 300 labeled examples in the CNN settings, 3646 in ViT-DomainNet and 1332 in ViT-CIFAR-20.

Every run uses the same weights-only grid. Bit widths are $\{2, 3, 4, 5, 6, 8\}$ and clipping percentiles $\{99.0, 99.5, 100.0\}$, crossed with per-tensor or per-channel scaling and an endpoint option for the first and last quantizable layers, giving $6 \times 3 \times 2 \times 2 = 72$ candidates. Clipping percentiles come from the weight tensors, so candidate construction consumes no target data. Coefficients are chosen by cross-validation within $\mathcal{S}$, five folds when $n \geq 25$ and leave-one-out below. The distortion term is evaluated once on the complete unlabeled target pool at every budget; only the supervised term uses the labeled subsample of size n . Supplement B states the full protocol, including fold construction and seeds. Probabilities are lower-clipped at 10^{-8} in the CNN pipelines and 10^{-12} in the ViT pipelines, following each implementation; we verify that the choice of floor changes regret by less than 0.001 across three floors spanning six orders of magnitude, so the magnitudes reflect the candidate families and not the clipping.

Selectors. The three members of (1) defined in Section 3 are evaluated against two references that spend the same labels: distortion alone, which minimizes D^{KL} over the target examples available at that budget, and direct validation, which minimizes empirical cross-entropy on $\mathcal{S}$. The teacher component of the alignment statistic is evaluated as a further ablation. Full definitions are in Supplement Table S3.

Selection without Target Labels

Minimum KL distortion is close to a constant rule: on the complete pool it chooses the same configuration in all 134 candidate families, 8-bit, per-channel, unclipped, with the first and last quantizable layers in full precision. We write $\max_g L_{\text{test}}(g) - \min_g L_{\text{test}}(g)$ for a family’s *cross-entropy spread*, the regret of choosing its worst member, which differs by a factor of five across our settings. Distortion’s mean regret is 0.220 on ResNet50 and 0.087 on MobileNetV2 over the pooled executions, so the label-free question is whether the least-compressed candidate is the empirical cross-entropy minimizer, reproducible with no target data at all. The choice is also near-degenerate: the two 8-bit per-channel unclipped candidates differing only in the endpoint option are separated

Table 1: Selection regret, mean $\pm$ standard deviation over runs. Columns are the four architecture–dataset settings: ResNet50 (RN50) and MobileNetV2 (MNV2) on DomainNet with 30 classes, and Vision Transformers on DomainNet with 50 classes (ViT-DN) and on held-out-domain CIFAR-20 with 20 classes (ViT-C20). *Every supervised selector here uses $n = 10$ labels*, the smallest budget we evaluate, so all columns are comparable; each gets the same ten examples, including those its coefficient selection consumes. The label-free rules use none, so their values hold at any n . Bold-face marks the lowest mean per column. Row labels give the venue and year; the corresponding references are Hendrycks and Gimpel (2017) for average confidence and entropy, Garg et al. (2022) for ATC, Deng et al. (2023) for nuclear norm, Guillory et al. (2021) for DOC, Tu et al. (2024) for SoftmaxCorr, Lu et al. (2023) for COT and Liu et al. (2023) for distortion. Runs are 45 per CNN architecture over three executions, 17 and 27 on the ViT settings; the other budgets are in Supplement Table S9.

Method	RN50	MNV2	ViT-DN	ViT-C20
Direct validation	0.182 $\pm$ 0.06	0.227 $\pm$ 0.13	0.208 $\pm$ 0.12	0.222 $\pm$ 0.10
CE-combo (ours)	0.166 $\pm$ 0.09	0.103 $\pm$ 0.07	0.139 $\pm$ 0.14	0.115 $\pm$ 0.06
Alignment (ours)	0.168 $\pm$ 0.10	0.088 $\pm$ 0.07	0.205 $\pm$ 0.33	0.150 $\pm$ 0.06
SoftmaxCorr [TMLR’24]	0.212 $\pm$ 0.18	0.061 $\pm$ 0.27	0.224 $\pm$ 0.47	0.083 $\pm$ 0.06
COT [NeurIPS’23]	0.257 $\pm$ 0.21	0.007 $\pm$ 0.01	0.278 $\pm$ 0.66	0.098 $\pm$ 0.05
Nuclear norm [ICML’23]	0.277 $\pm$ 0.20	0.013 $\pm$ 0.03	0.280 $\pm$ 0.66	0.098 $\pm$ 0.05
DOC [ICCV’21]	0.221 $\pm$ 0.20	0.086 $\pm$ 0.08	0.284 $\pm$ 0.69	0.095 $\pm$ 0.04
Distortion [CVPR’23] (anch.)	0.220 $\pm$ 0.20	0.087 $\pm$ 0.08	0.284 $\pm$ 0.69	0.096 $\pm$ 0.04
ATC-MC [ICLR’22]	14.489 $\pm$ 3.05	4.461 $\pm$ 2.01	0.284 $\pm$ 0.69	0.095 $\pm$ 0.04
ATC-NE [ICLR’22]	15.125 $\pm$ 0.31	3.985 $\pm$ 2.31	0.284 $\pm$ 0.69	0.095 $\pm$ 0.04
Entropy [ICLR’17]	15.125 $\pm$ 0.31	10.633 $\pm$ 1.57	0.792 $\pm$ 1.17	0.572 $\pm$ 0.62
Avg. confidence [ICLR’17]	15.125 $\pm$ 0.31	10.654 $\pm$ 1.53	0.919 $\pm$ 1.47	0.442 $\pm$ 0.62

by a distortion gap of order 10^{-4} , their ordering reverses on a 150-example half-pool in one of five shifts in two of the six CNN executions, and their target cross-entropies differ by 0.0015.

Table 1 compares every selector at the smallest budget, so no row is advantaged by labels another does not see. Regret is the mean over runs. The two blocks answer whether spending n labels beats the best free option; paired tests are between supervised selectors (Supplement Table S5).

We screen the state of the art in label-free performance estimation, spanning its four families: average confidence and entropy (Hendrycks and Gimpel 2017), the two ATC variants (Garg et al. 2022), nuclear norm (Deng et al. 2023), DOC (Guillory et al. 2021) and SoftmaxCorr (Tu et al. 2024), COT (Lu et al. 2023), and teacher distortion (Liu et al. 2023). All nine are computed from predictive distributions alone. ATC and DOC need a calibration set, which we form with the teacher’s arg-max rather than with labels, so the two rows are teacher-pseudo-label variants of the published methods (Supplement B). Direct validation cannot use the unlabeled inputs, so the natural missing comparison is a semi-supervised criterion using both (Shanmugam et al. 2025); we do not evaluate one, and our margin over direct validation should be read with that in mind.

The confidence-based group fails by two orders of magnitude: on ResNet50 average confidence and entropy essentially select the worst candidate, and the two ATC variants select from the same collapsed region. Figure 1 shows why: low-bit quantization produces degenerate predictors at chance accuracy whose confidence *exceeds* that of the candidates minimizing target cross-entropy, 1.000 against 0.408 on ResNet50. Any statistic reading confidence as correctness therefore orders the family close to backwards. The two ViT settings invert the pattern: their degenerate candidates stay under-confident and the same estimators select well, so the cause is the degradation regime and not the estimator. Supplement C shows the magnitudes do not depend on probability clipping, and Supplement Table S6 that the same failure appears when estimators are scored on accuracy. The diagnostics that distinguish the two regimes in our experiments are measurable only with target labels.

The output-distribution group is a different matter, and the multiple-comparison correction sharpens it. On ResNet50 we cannot distinguish any of the four from the anchor with the available clustered sample, and all four incur substantial mean regret. On MobileNetV2 three of them have clearly lower mean regret than the anchor. Teacher distortion is therefore not the strongest label-free rule available, and we do not claim it is; what it offers is stability under the degradation regime that defeats the confidence group, which makes it usable as a shrinkage target rather than as a selector.

Comparing the two blocks of Table 1 answers whether labels are worth acquiring, and the answer differs sharply by architecture. On ResNet50 anchoring reduces mean regret from 0.166 to 0.007 as the budget grows, against 0.212 for the best label-free rule. On MobileNetV2 a label-free rule wins at every budget up to the complete pool: COT attains 0.007 out of a cross-entropy spread of 11.4, and no supervised selector improves on it until $n = 300$. The contrast is consistent with the headroom the best label-free rule leaves, which differs by a factor of twenty between the two, 1.4% of the attainable damage against 0.065%. Labels buy selection accuracy where the label-free problem is unsolved, a property of the candidate family, not of the selector.

Distortion is immune to that failure yet still leaves regret on the table. A post-hoc audit (Table 2) shows that on ResNet50 the candidate chosen by validation cross-entropy has lower calibration error and cross-entropy but lower accuracy: paired run-level differences are -0.122 [$-0.161, -0.082$] in ECE and -0.208 [$-0.310, -0.113$] in cross-entropy. Calibration, not accuracy, is the dominant source of disagreement, and this is what a 0–1 loss cannot recover. Supplement C checks that the difficulty is not an artifact of one catastrophic candidate by restricting the family.

Selection with Scarce Clean Labels

Spending a small number of target labels changes the picture. Table 3 reports median regret across budgets for the three selectors.

With ten labels CE-combo improves on direct validation in *all four* settings. Across the twenty-three comparisons of the budget ablation (Supplement Table S9) it wins in eleven and loses in twelve, strongly budget-dependent: four of four at ten

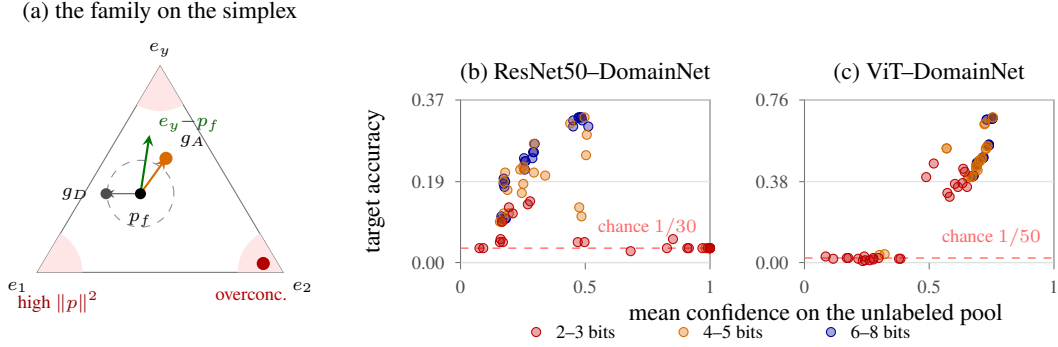


Figure 1: Why the label-free rule needs help, and why confidence cannot supply it. (a) Two candidates on a level set of the teacher distortion, g_D and g_A , are indistinguishable to the label-free rule; the supervised term breaks the tie, and alignment prefers movement along $e_y - p_f$, while an overconcentrated candidate sits near a vertex, confident and far from the teacher. (b, c) Each point is one of the 72 candidates of a single run, positioned by its mean confidence on the unlabeled pool and its target accuracy, coloured by bit width, with the dashed line at chance. On ResNet50 confidence and accuracy are negatively related across the family ($\rho = -0.32$ in this run); on ViT-DomainNet the relation is strongly positive ($\rho = +0.93$), which is why the same estimators select well there.

Table 2: Post-hoc calibration audit, computed on the first execution of each CNN setting. Upper block: medians over its 15 runs per architecture, ECE with 15 equal-width bins, accuracy top-1. Lower block: mean run-level differences (validation CE minus distortion) with run-bootstrap 95% intervals, leading zeros omitted. Test labels are used only for explanation.

Setting	Selected by	ECE	Acc.	CE
ResNet50	Distortion	0.216	0.332	2.865
	Validation CE	0.083	0.327	2.745
MobileNetV2	Distortion	0.227	0.308	3.019
	Validation CE	0.195	0.308	2.884
<i>Paired difference, validation CE minus distortion</i>				
ResNet50		-.122	-.042	-.208
95% CI		[-.161, -.082]	[-.056, -.027]	[-.310, -.113]
MobileNetV2		-.013	+.006	-.084
95% CI		[-.019, -.007]	[+.002, +.009]	[-.121, -.049]

labels, three of four at twenty-five, and four of fifteen from fifty onward. At ten labels the mean regret falls from 0.227 to 0.103 on MobileNetV2, from 0.222 to 0.116 on ViT-CIFAR-20, from 0.208 to 0.140 on ViT-DomainNet and from 0.182 to 0.166 on ResNet50. On ResNet50 the median does not fall, so the gain there comes from removing severe failures rather than from shifting the typical outcome. From fifty labels onward it loses in eleven of fifteen comparisons, so its stabilizing benefit usually stops compensating for anchor bias once the supervised estimate is precise. The value of anchoring is a small-budget effect.

Anchoring also protects the upper tail: at ten labels the 95th percentile of regret falls from 0.58 to 0.49 on ResNet50 and from 1.18 to 0.71 on MobileNetV2, and $Pr(R > 0.1)$ from 0.30 to 0.23 on the latter (Supplement Table S10). Paired run-level inference for both supervision regimes is in Table S5.

What the Directional Term Contributes

The directional term admits a control that isolates exactly what it claims to add. Permuting the observed labels before the statistic is formed leaves the marginal distribution and scale of A_S intact and destroys only the correspondence between labels and inputs, so the control keeps the shrinkage of (1) and none of its direction. It receives the identical coefficient grid and cross-validation.

Supplement Table S13 reports it. The control behaves as its construction predicts, though not exactly: its regret tracks pure distortion to within 0.020 across settings and budgets, with a mean absolute deviation of 0.006. The coefficient it selects is zero in 71% of ResNet50 and 76% of MobileNetV2 runs, median zero in both, so with the correspondence destroyed the procedure usually reverts to the anchor, and the residual deviation comes from the runs where it does not. That makes the comparison informative only where reverting to the anchor is costly.

ResNet50 is that case, and there the direction contributes throughout: alignment beats the control at every budget, from 0.1870 against 0.2170 at ten labels to 0.0270 against 0.2320 at 150, the gap widening as labels accumulate. In the other three settings distortion is already within a few hundredths of the optimum, so reverting to it forfeits little and the control is not separated from alignment. The directional term therefore earns its place conditionally. Relating the gain from anchoring to the regret of pure distortion across the nineteen setting-budget cells beyond the smallest budget shows the condition is not a property of the anchor: the association is weak and of the wrong sign (Spearman -0.25), which we read as a descriptive diagnostic, not a formal test, since the cells are not independent. What governs the gain is the label budget. With ten labels anchoring improves on direct validation in all four settings, and from fifty labels onward it loses in three of them. That is what a shrinkage rule should do: once the supervised estimate is reliable, the anchor stops helping.

Table 3: Median target cross-entropy selection regret with clean labels, over run-labeled-subset outcomes. Dist. is label-free and constant across budgets. Boldface marks the lowest median among the supervised selectors in each row. The largest budget is the largest evaluated in each setting, not the size of its labeled pool. Runs are 45 per CNN architecture, pooling three end-to-end executions, and 17 and 27 on the two ViT settings.

Setting	n	Dist.	Val-CE	CE-combo	Align
ResNet50–DomainNet	10	0.2323	0.1756	0.1941	0.1870
	25	0.2323	0.0972	0.0948	0.1143
	50	0.2323	0.0571	0.0498	0.0635
	100	0.2323	0.0267	0.0271	0.0342
	150	0.2323	0.0232	0.0239	0.0270
	300	0.2323	0.0000	0.0012	0.0024
MobileNetV2–DomainNet	10	0.0778	0.2479	0.1055	0.0816
	25	0.0778	0.0644	0.0349	0.0399
	50	0.0778	0.0194	0.0221	0.0241
	100	0.0778	0.0073	0.0046	0.0049
	150	0.0778	0.0024	0.0033	0.0026
	300	0.0778	0.0000	0.0000	0.0000
ViT–DomainNet	10	0.0200	0.1722	0.0725	0.0501
	25	0.0200	0.0746	0.0445	0.0434
	50	0.0200	0.0482	0.0207	0.0295
	100	0.0200	0.0188	0.0092	0.0183
	150	0.0200	0.0151	0.0082	0.0110
ViT–CIFAR-20	10	0.1009	0.1938	0.0990	0.1325
	25	0.1009	0.0952	0.0856	0.1010
	50	0.1009	0.0436	0.0638	0.0764
	100	0.1009	0.0352	0.0483	0.0715
	150	0.1009	0.0263	0.0421	0.0531
	200	0.1009	0.0219	0.0382	0.0458

The two components behave heterogeneously across settings rather than along one axis. Retaining only the teacher component, whose statistic uses no observed class, is the best of the three variants on ResNet50 at ten labels (0.145) and the worst on the other three settings at that budget. Alignment is thus not a mechanism of uniform value. It is a correction whose value varies across settings, and anchor regret does not predict that variation; the label budget is the clearer moderator.

Selection under Symmetric Label Corruption

Each sampled label is retained with probability $1 - \eta$ and otherwise replaced uniformly by one of the $K - 1$ incorrect classes, so a corrupted label always differs from the clean one. Corruption changes only the labels supplied to the selectors, never the candidates or test set.

Table S12 shows that anchoring becomes more valuable as supervision degrades. At $\eta = 0.4$ both anchored selectors improve on direct validation on both architectures. The directional variant attains the lowest median in each case, reducing regret from 0.165 to 0.102 on ResNet50 and from 0.779 to 0.393 on MobileNetV2.

Scarce and Corrupted Labels Together

The experiments so far vary one factor at a time, leaving out the regime a practitioner is most likely to face. Table 4 crosses the two, giving every selector the same n labels at the same rate and corrupting those its coefficient selection consumes as well.

Supplement C shows that neither the input set on which the anchor is evaluated nor its identity as the teacher is essential to what follows. Anchoring helps far more here than with clean labels. Across the twenty cells with $\eta > 0$ CE-combo improves on direct validation in every one, against eleven of twenty-three in the clean sweep. The margins are large where both scarcity and corruption bite: at ten labels and $\eta = 0.4$ mean regret falls from 0.393 to 0.278 on ResNet50 and from 0.864 to 0.485 on MobileNetV2. Corruption inflates the variance of the supervised estimate, and that is precisely when shrinkage should pay. Here it pays at every budget, and not merely the smallest.

Validation accuracy behaves as the attenuation identity predicts and thereby marks the limit of the anchored rule. At ten labels it is catastrophic, above 1.1 on MobileNetV2, since a 30-class accuracy estimate from ten examples is almost uninformative. But its ordering degrades only by a common factor, so once the estimate is precise it becomes the most corruption-tolerant of the three: on MobileNetV2 it attains the lowest regret in all six cells with $n \geq 50$ and $\eta > 0$, reaching 0.028 at $n = 300$ and $\eta = 0.4$ where CE-combo reaches 0.803. This points to a division of labour: anchor when labels are few, and consider a bounded accuracy criterion when they are plentiful but unreliable, a pattern we observe on MobileNetV2 and not on ResNet50.

Does Selection Survive a Cost Constraint?

Minimizing predictive cross-entropy over the whole family answers a question a practitioner would not ask alone, since candidates differ in deployment cost. We repeat the exercise as a constrained problem, $\min_g L(g)$ subject to $C(g) \leq B$, with C the weight memory relative to full precision, with the frontier in Supplement Table S14.

Supplement Table S15 applies the selectors inside the constraint, with the candidate set, the labeled sample, the coefficient search and the oracle all restricted to $\mathcal{G}_B$. The small-budget pattern survives: at ten labels CE-combo improves on direct validation in seven of the eight budget–architecture cells and loses in one, the tightest budget on MobileNetV2, where the admissible family is small enough that the choice barely matters. Restricting the problem neither removes the advantage of anchoring at small budgets nor creates one at large budgets.

Two effects separate. The price of the constraint is strongly architecture-dependent: on ResNet50 the frontier is nearly flat above a fifth of full precision and $\leq 22\%$ costs 0.007, while on MobileNetV2 it is steep everywhere and $\leq 22\%$ already costs 0.868. The effect that matters for selection survives intact: within every budget the spread between the worst and best admissible candidate stays above 10 in cross-entropy, essentially unchanged. Constraining cost removes the most expensive candidates but does not make the re-

Table 4: Label budget crossed with corruption rate: mean target cross-entropy regret over 15 runs per architecture and ten corruption realizations per cell. Unlike the clean-label tables this grid is computed on the first complete execution of each setting, since crossing five budgets with three corruption rates and ten realizations makes pooling all three executions prohibitively slow. Every selector receives the same n labels at the same rate η , and the labels its coefficient selection consumes are corrupted too. Boldface marks the lowest mean in each cell.

Setting	n	$\eta = 0$			$\eta = 0.2$			$\eta = 0.4$		
		Val-CE	Val-Acc	CE-combo	Val-CE	Val-Acc	CE-combo	Val-CE	Val-Acc	CE-combo
ResNet50–DomainNet	10	0.168	0.630	0.159	0.326	1.497	0.234	0.393	2.485	0.278
	25	0.100	0.249	0.091	0.145	0.427	0.111	0.361	0.792	0.230
	50	0.051	0.262	0.053	0.118	0.269	0.083	0.317	0.262	0.216
	100	0.025	0.227	0.030	0.104	0.239	0.071	0.288	0.240	0.194
	300	0.008	0.219	0.010	0.087	0.231	0.073	0.232	0.240	0.202
MobileNetV2–DomainNet	10	0.285	1.140	0.127	0.728	2.021	0.406	0.864	2.219	0.485
	25	0.107	0.201	0.046	0.308	0.464	0.143	0.708	0.851	0.513
	50	0.048	0.054	0.027	0.285	0.079	0.135	0.765	0.230	0.601
	100	0.015	0.044	0.008	0.260	0.037	0.136	0.768	0.036	0.684
	300	0.001	0.018	0.001	0.193	0.019	0.099	0.844	0.028	0.803

Table 5: Empirical validation of the symmetric-corruption attenuation identity at the complete labeled pool, pooling all three executions, 45 runs per architecture and ten corruption realizations per rate. Entries are through-origin slopes relating the corrupted quantity to its clean counterpart. The teacher component is unchanged to numerical precision at every rate.

Setting	η	Predicted	Label comp.	Val. accuracy
ResNet50–DomainNet	10%	0.897	0.881	0.881
	20%	0.793	0.800	0.802
	30%	0.690	0.683	0.687
	40%	0.586	0.577	0.579
MobileNetV2–DomainNet	10%	0.897	0.882	0.882
	20%	0.793	0.803	0.802
	30%	0.690	0.688	0.689
	40%	0.586	0.575	0.575

mainder interchangeable. The problem we study therefore survives a budget instead of being dissolved by it.

Does the Theory Predict the Measurement?

The attenuation result makes a numerical prediction. At the largest budget and $\eta \in \{0, 0.1, 0.2, 0.3, 0.4\}$ we regress the corrupted observed-label component through the origin on its clean counterpart, across candidates, subsets and runs; the predicted slope is $a_\eta = 1 - \frac{K}{K-1}\eta$. We repeat it for mean-centred validation accuracy, whose predicted slope is the same, since the derivation uses only linearity in the label indicator.

Table 5 reports both, with a maximum absolute discrepancy of 0.016 at the lowest corruption rate and below 0.011 elsewhere. Two consequences follow. The corruption implementation does what the analysis says, so the corruption results measure the intended quantity. And validation accuracy attenuates by the same factor as the Brier alignment, so the identity holds for the whole class and not for one statistic

alone, which removes attenuation as a reason to prefer alignment over noisy validation accuracy. It is not a performance guarantee.

With clean labels the coefficient has median 7.5 on ResNet50 and 3.5 on MobileNetV2, reverting to zero in 11% and 29% of runs; under 40% corruption on MobileNetV2 it reaches the grid maximum in 53%, so the grid binds there; Supplement Table S11 and Supplement Figure 3 give its distribution and the underlying regret curve. The frequency with which the procedure reverts to pure distortion is highest where the labeled signal is least useful, so the coefficient does the work the construction intends rather than sitting at a grid boundary.

5 Conclusion

Selecting among fixed quantized variants of a shared teacher is a distinct problem from building them, and today’s estimators do not solve it. Minimum teacher distortion selects the same eight-bit configuration in all 134 candidate families, and that candidate never minimizes target cross-entropy. Confidence-based estimators do worse than uninformative: where quantization produces degenerate predictors more confident than the healthy ones they select from the wrong end of the family, and no estimator we evaluate detects which regime holds.

Anchoring converts that stability into a usable selector. With ten target labels it reduces mean regret below direct validation in all four settings, and under corruption in every one of the twenty budget–rate cells. The identities show the members differ only in which function of the labeled sample is added, and the attenuation result gives the class of criteria whose label signal decays at one rate. Neither bounds selection regret, which we establish empirically.

The results also delimit the method. Anchoring is a general construction, not a property of the teacher: a stronger anchor yields a better anchored selector. And where labels are plentiful but unreliable a bounded accuracy criterion can do better, as on MobileNetV2.

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Supplementary Material

Table S1 collects the notation used throughout the paper and this supplement.

This supplement is organised as follows, with the part of the main paper each section supports. Supplement A states and proves the two formal results summarised in Section 3 and delimits what they do and do not bound. Supplement B gives the full experimental protocol behind Section 4, including the candidate grid, the selector definitions, the coefficient search stated as Algorithm 1, the seeding scheme and the run accounting. Supplement C reports the calibration audit and the candidate-family sensitivity checks that accompany Table 2. Supplement D collects the tables the main text cites but does not reproduce, together with the additional analyses that accompany them. Supplement E quantifies the two degradation regimes over all executions and is the source of the confidence figures quoted in Section 4. Supplement F documents the runs excluded from the final cohort.

A Score Identities and Theoretical Boundaries

Smoothness interpretation

Let $\delta_g^z(x) = z_g(x) - z_f(x)$. If $\ell(\cdot, y)$ has an L -Lipschitz gradient along the segment joining $z_f(x)$ and $z_g(x)$, the descent lemma gives $\ell(z_g(x), y) - \ell(z_f(x), y) \leq \langle \nabla_z \ell(z_f(x), y), \delta_g^z(x) \rangle + \frac{L}{2} \|\delta_g^z(x)\|_2^2$. With $a_\ell(x, y; g, f) = -\langle \nabla_z \ell(z_f(x), y), \delta_g^z(x) \rangle$, averaging over $\mathcal{S}$ gives

$$\widehat{L}_\mathcal{S}(g) - \widehat{L}_\mathcal{S}(f) \leq -A_{\ell, \mathcal{S}}(g, f) + \frac{L}{2|\mathcal{S}|} \sum_{(x, y) \in \mathcal{S}} \|\delta_g^z(x)\|_2^2$$

For probability outputs, pointwise Pinsker control gives $\|\delta_g^p(x)\|_2^2 \leq \|\delta_g^p(x)\|_1^2 \leq 2 \text{KL}(p_g(\cdot | x) \| p_f(\cdot | x))$, which provides direct empirical control when the KL and perturbation terms are averaged over the same inputs. Our implemented selector averages the distortion over the whole pool and the supervised term over $\mathcal{S}$, so the bound applies to the restriction of both terms to a common input set and not to the score as evaluated. We present it as a local structural interpretation rather than as practical control: the Lipschitz constant of the cross-entropy gradient in probability space grows with the inverse of the probability floor, so the resulting bound is numerically weak at the floors we use.

Exact quadratic identity

Proposition 1 (Quadratic identity) Write $\delta_g^p(x) = p_g(x) - p_f(x)$, and on a labeled set $\mathcal{S}$ let $D_\mathcal{S}^{(2)}(g, f) = |\mathcal{S}|^{-1} \sum \|\delta_g^p(x)\|_2^2$ and $A_\mathcal{S}^B(g, f) = |\mathcal{S}|^{-1} \sum \langle \delta_g^p(x), e_y - p_f(x) \rangle$. If both terms are averaged over the same $\mathcal{S}$, then

$$D_\mathcal{S}^{(2)}(g, f) - 2A_\mathcal{S}^B(g, f) = \widehat{\text{Br}}_\mathcal{S}(g) - \widehat{\text{Br}}_\mathcal{S}(f)$$

Proof. Every labeled example satisfies $\|p_g(x) - e_y\|_2^2 - \|p_f(x) - e_y\|_2^2 = \|\delta_g^p(x)\|_2^2 - 2\langle \delta_g^p(x), e_y - p_f(x) \rangle$; averaging over $\mathcal{S}$ gives the claim. $\square$

Evaluating both sides candidate by candidate over the 90 CNN runs, the two agree to 1.3×10^{-15} , that is, to numerical

precision. The teacher term is candidate-independent, so with $\lambda = 2$ and both terms on the same inputs the criterion selects exactly the same candidate as empirical Brier validation. This is an identity between selection criteria, presented as a consistency check rather than a new scoring rule.

Observed-label and teacher components

By linearity, $A_\mathcal{S}^B = A_\mathcal{S}^{\text{lab}} + A_\mathcal{S}^{\text{teach}}$ with $A_\mathcal{S}^{\text{lab}} = |\mathcal{S}|^{-1} \sum \langle \delta_g^p(x), e_y \rangle$ and $A_\mathcal{S}^{\text{teach}} = -|\mathcal{S}|^{-1} \sum \langle \delta_g^p(x), p_f(x) \rangle$. For any common input set $\mathcal{V}$,

$$\begin{aligned} D_\mathcal{V}^{(2)}(g, f) - 2A_\mathcal{V}^{\text{teach}}(g, f) \\ = \frac{1}{|\mathcal{V}|} \sum_{x \in \mathcal{V}} [\|p_g(x)\|_2^2 - \|p_f(x)\|_2^2] \end{aligned}$$

so minimizing the canonical fully quadratic teacher-only criterion favors the candidate with the smallest average squared predictive norm. Taking $\mathcal{V} = \mathcal{U}$ gives a label-free predictive-concentration criterion. This does not imply that the empirical KL-based teacher-component ablation is identical to concentration ranking, since that ablation retains KL distortion and a selected coefficient.

Attenuation under symmetric corruption

Proposition 2 (Common attenuation) Let a label-dependent criterion be $a_g(x)^\top e_y$ and let each label be replaced, with probability η , by a class drawn uniformly from the $K - 1$ incorrect ones. Then

$$\mathbb{E}[a_g^\top e_{\tilde{y}} | y] = \left(1 - \frac{K}{K-1}\eta\right) a_g^\top e_y + \frac{\eta}{K-1} \mathbf{1}^\top a_g$$

If $\mathbf{1}^\top a_g$ does not vary across candidates, the second term is a common offset and candidate comparisons are multiplied by the single factor $1 - \frac{K}{K-1}\eta$. The condition holds for teacher contrasts, whose coordinates sum to zero, and for accuracy indicators, whose coordinates sum to one; it fails for cross-entropy, whose coefficient vector $-\log p_g$ has a candidate-dependent coordinate sum.

Proof. Conditional on the clean class y , $\mathbb{E}[e_{\tilde{y}} | y] = (1 - \frac{K}{K-1}\eta)e_y + \frac{\eta}{K-1} \mathbf{1}$; taking the inner product with a_g gives the display. $\square$

Applying Proposition 2 to the alignment statistic, where $a_g = p_g(x) - p_f(x)$ so that $\langle a_g, \mathbf{1} \rangle = 0$,

$$\mathbb{E}[A_\mathcal{S}^{\text{noisy}} | \mathcal{S}] = \left(1 - \frac{K}{K-1}\eta\right) A_\mathcal{S}^{\text{lab}} + A_\mathcal{S}^{\text{teach}}$$

The factor is positive whenever $\eta < (K - 1)/K$; for $K = 30$ this bound is $29/30 \approx 0.967$, outside the range we evaluate, so within our experiments the label signal shrinks by a known factor and never inverts. Both slopes are verified empirically in Table 5, to within 0.016.

The identity is not specific to Brier alignment. The derivation uses only that $A_\mathcal{S}^{\text{lab}}$ is linear in e_y and that the candidate-teacher difference is orthogonal to $\mathbf{1}$. Any functional linear in e_y obeys the same attenuation, including empirical accuracy:

$$\mathbb{E}[\widehat{\text{Acc}}^{\text{noisy}}(g)] = \left(1 - \frac{K}{K-1}\eta\right) \widehat{\text{Acc}}(g) + \frac{\eta}{K-1}$$

Table S1: Notation used in the framework and the supplementary analyses.

Symbol	Meaning	Symbol	Meaning
$f, p_f(\cdot x)$	fixed teacher and its predictive distribution	$g \in \mathcal{G}, p_g(\cdot x)$	candidate and its predictive distribution
$\mathcal{G}, M$	fixed candidate family and its cardinality	$\mathcal{U}$	unlabeled target input pool
$\mathcal{S}, n$	labeled target sample and its size	$\mathcal{T}$	disjoint target test set, evaluation only
e_y	one-hot vector for class y	$z_f(x), z_g(x)$	teacher and candidate outputs in a chosen representation
$\delta_g^p(x)$	probability change $p_g(x) - p_f(x)$	$\Phi_{\mathcal{S}}(g)$	supervised term added to the distortion
$D_V^d(g, f)$	teacher distance of type d on input set $\mathcal{V}$	$a_{\ell}(x, y; g, f)$	per-example alignment contribution for loss ℓ
$A_{\mathcal{S}}(g, f)$	alignment on $\mathcal{S}$	$\widehat{\text{Br}}_{\mathcal{S}}$	empirical Brier score on $\mathcal{S}$
$A_{\mathcal{S}}^{\text{lab}}, A_{\mathcal{S}}^{\text{teach}}$	observed-label and teacher components	$A_{\mathcal{S}}^{\pi}$	permutation control, residual directions $e_y - p_f(x)$ permuted within $\mathcal{S}$
α	weight of the supervised term in (1)	K	number of classes
$\lambda, \lambda_{\max}$	weight of the directional term and the largest value in its grid	$\delta_g^z(x)$	output change $z_g(x) - z_f(x)$ in the chosen representation
$L_{\text{test}}(g)$	held-out target cross-entropy	$R(\hat{g})$	cross-entropy selection regret
η	label-corruption probability	$C(g)$	deployment cost of candidate g
F	number of cross-validation folds	$\mathcal{G}_B$	candidates admissible under budget B

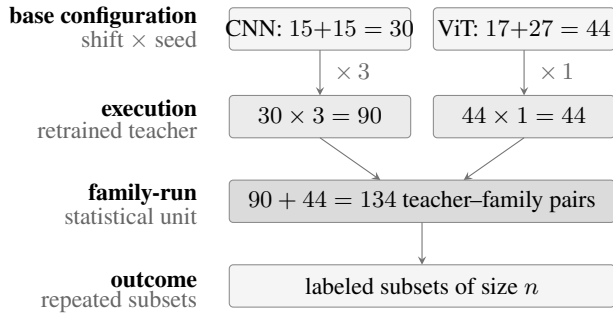


Figure 2: Levels of the experimental design. A base configuration is a shift–seed pair: 30 on the two CNN settings and 44 on the two ViT settings, 74 in all. Each CNN configuration is executed three times with a retrained teacher on identical splits and each ViT configuration once, so the number of independently trained teachers, and hence of *family-runs*, is $30 \times 3 + 44 \times 1 = 134$: 45 per CNN architecture, 17 on ViT–DomainNet and 27 on ViT–CIFAR-20. The family-run is the statistical unit: subset outcomes are averaged within it. For paired inference the three executions of a configuration are averaged as well, leaving 15 paired units per CNN architecture.

affine in clean accuracy with the same positive slope, so noisy validation accuracy preserves the clean accuracy ordering in expectation. Table 5 verifies both slopes on the same runs.

B Experimental Protocol

Runs, executions, and resampling units

Four nested levels appear throughout, and Supplement Figure 2 shows how they relate. Table S2 distinguishes independent candidate-family runs from within-run repetitions and from whole-pipeline re-executions.

Clarification of the main-text summary. The figure 3646 quoted in Section 4 is the pool size for one ViT–DomainNet shift; across the complete 17-run cohort the selection-pool size varies from 2526 to 6308. Each run uses its own complete

Table S2: Accounting of the experimental design. A run contains one teacher and its fixed 72-candidate family. Executions are independent end-to-end repetitions with retrained teachers on identical splits. Results for the two CNN settings pool all three executions, giving 45 runs per architecture; the two ViT settings have one execution each. [†]for $n < 300$; the complete pool is evaluated once at $n = 300$. On the two ViT settings every clean budget is below the pool size, so all of them use the full 25 repetitions. The counts in this table are those of the budget sweep; the budget-by-rate grid of the main text draws ten repetitions per cell at every rate, including $\eta = 0$. The ViT–DomainNet selection pool varies by shift, from 2526 to 6308 examples; the largest supervised budget remains $n = 150$.

Setting	Cfg.	Ex.	Runs	Pool	Clean	Noise
RN50–DN	15	3	45	150+150	25 [†]	10
MNV2–DN	15	3	45	150+150	25 [†]	10
ViT–DN	17	1	17	2526–6308	25	10
ViT–C20	27	1	27	666+666	25	10
Total	74	–	134	–	–	–

pool, and the largest supervised budget is fixed at $n = 150$ throughout.

Candidate construction. Quantization is weight-only and simulated in floating point, so no integer kernel is required. For a weight tensor w , bit width b , and clipping percentile q , the scale is

$$s = \frac{\text{pct}_q(|w|)}{2^{b-1} - 1}$$

We use the symmetric signed range $[-(2^{b-1} - 1), 2^{b-1} - 1]$, zero point zero, and round-half-to-even. Weights are clipped to $\pm \text{pct}_q(|w|)$ before scaling. Per-tensor quantization uses one scale for the complete tensor. Per-channel quantization reduces over all axes except the last and uses one scale per output channel. Only kernel weights are quantized; biases, normalization parameters, and embeddings remain in floating point. When endpoint skipping is enabled, the first and

last quantizable layers also remain in floating point.

Write the ordered arrays $B = (2, 3, 4, 5, 6, 8)$, $Q = (99.0, 99.5, 100.0)$ and $G = (\text{tensor}, \text{channel})$, and let

$$\begin{aligned} \mathcal{B} &= \{2, 3, 4, 5, 6, 8\}, & \mathcal{Q} &= \{99.0, 99.5, 100.0\}, \\ \mathcal{G} &= \{\text{tensor}, \text{channel}\}, & \mathcal{E} &= \{0, 1\} \end{aligned}$$

The candidate family is $\mathcal{B} \times \mathcal{Q} \times \mathcal{G} \times \mathcal{E}$, giving $6 \times 3 \times 2 \times 2 = 72$ candidates per teacher. The endpoint indicator varies fastest, followed by granularity, clipping percentile, and bit width. For zero-based index i ,

$$\begin{aligned} b_i &= B_{\lfloor i/12 \rfloor}, & q_i &= Q_{\lfloor i/4 \rfloor \bmod 3}, \\ g_i &= G_{\lfloor i/2 \rfloor \bmod 2}, & e_i &= i \bmod 2 \end{aligned}$$

Here, $e_i = 1$ means that the first and last quantizable layers remain in floating point.

The selection pool is the union of the quantization-calibration split and the selection split. Because clipping percentiles are computed from the weight tensors rather than from data, the calibration split plays no role in constructing the candidate family, and its inclusion in the selection pool does not leak candidate-construction information. We report the composition explicitly because the two halves are not interchangeable: on a 150-example half-pool the distortion ordering between the two 8-bit per-channel unclipped candidates can reverse.

Test splits contain 900 examples per run for the DomainNet CNNs, 1824 for ViT-DomainNet and 668 for ViT-CIFAR-20. All test sets are fixed, disjoint from the selection pools, and used only after candidate selection.

Teacher training. CNN teachers: transfer learning from the framework’s ImageNet weights and its own preprocessing, five epochs of head training at learning rate 10^{-3} with the backbone frozen, then three unfrozen at 10^{-5} , both with Adam and batch size 64. ViT teachers: continual distillation over a vit_base_patch16_224 backbone (Dosovitskiy et al. 2021) at 224×224 , Adam at 10^{-4} , batch size 64, distillation temperatures 2.0 and 10.0. Seeds 0, 1 and 2 index the repetitions, each set once per run for the Python, NumPy and framework generators.

Estimator variants. SoftmaxCorr and COT are evaluated with a uniform class prior, which the balanced target splits permit; this is the variant reported throughout.

Calibration of the label-free estimators. Two of the nine need a calibration set: ATC learns a threshold and DOC a reference accuracy, both defined on labeled data in their original form. Our pipelines retain no source-domain predictions, so we calibrate them on the target calibration split using the teacher’s arg-max as the reference label, which keeps them label-free and uses the reference this setting already provides. One further estimator is outside the comparison for a structural reason: Li et al. (2026) requires an external foundation model as its reference, which changes the resource assumption of this setting, where the teacher is the reference.

Selector definitions and coefficient selection

Table S3 lists all selectors.

Algorithm 1: Anchored selection. The anchor is evaluated once on the complete pool and reused at every budget and fold; only the supervised term sees $\mathcal{S}$. Written for an added supervised term, as in CE-combo; the directional variants replace $c\phi$ by $-ca$ and are otherwise identical.

Require: teacher f , candidate family $\mathcal{G}$, unlabeled pool $\mathcal{U}$, labeled sample $\mathcal{S}$ of size n , coefficient grid $\mathcal{C}$ with $0 \in \mathcal{C}$, fold count F

Ensure: selected candidate $\hat{g}$

- 1: $D(g) \leftarrow \frac{1}{|\mathcal{U}|} \sum_{x \in \mathcal{U}} \text{KL}(p_f(\cdot | x) \| p_g(\cdot | x))$ for every $g \in \mathcal{G}$ {once per run; no labels}
- 2: $\phi(g, x, y) \leftarrow$ per-example supervised statistic
- 3: partition $\mathcal{S}$ into contiguous folds $\mathcal{S}_1, \dots, \mathcal{S}_F$; $F = 5$ if $n \geq 25$, else $F = \min(n, 10)$, which is leave-one-out at every budget we evaluate
- 4: **for** $c \in \mathcal{C}$ in increasing order **do**
- 5: $\ell(c) \leftarrow 0$
- 6: **for** $k = 1$ **to** F **do**
- 7: $\hat{g}_{c,-k} \leftarrow \arg \min_{g \in \mathcal{G}} D(g) + c \frac{1}{|\mathcal{S}_{-k}|} \sum_{(x,y) \in \mathcal{S}_{-k}} \phi(g, x, y)$
- 8: $\ell(c) \leftarrow \ell(c) + \frac{1}{|\mathcal{S}_k|} \sum_{(x,y) \in \mathcal{S}_k} -\log p_{\hat{g}_{c,-k}}(y | x)$
- 9: **end for**
- 10: **end for**
- 11: $\hat{c} \leftarrow \arg \min_{c \in \mathcal{C}} \ell(c)$, ties to the first grid value
- 12: **return** $\hat{g} \leftarrow \arg \min_{g \in \mathcal{G}} D(g) + \hat{c} \frac{1}{n} \sum_{(x,y) \in \mathcal{S}} \phi(g, x, y)$

Implementation convention. In all principal experiments $D_{\mathcal{U}}^{\text{KL}}$ is computed on the complete selection-input pool and remains fixed across label budgets and cross-validation folds; only the supervised component is evaluated on $\mathcal{S}$. The same- $\mathcal{S}$ construction appears only in the quadratic identity of Proposition 1 and in the anchor-placement sensitivity analysis of Supplement C. No candidate is retrained or adapted at any point, and the held-out test set is never touched. Five folds are used when $n \geq 25$ and leave-one-out below; folds are contiguous blocks of the labeled sample in its stored order and become singletons when $n < F$. Every draw that varies within a run is seeded deterministically from the quantity that indexes it, so a replication recovers the same labeled subsets and the same corruption realizations without storing them: the subset of size n at repetition r uses seed $10^5 + 7919n + r$, and the corruption realization at rate η and repetition r uses seed $9500 + \lfloor 100\eta \rfloor + r$, each initializing a PCG64 generator. Write $\mathcal{S} = \bigcup_{k=1}^F \mathcal{S}_k$ for the fold partition and $\mathcal{S}_{-k}$ for its complement. For a coefficient c in the grid $\mathcal{C}$, let $\hat{g}_{c,-k}$ be the candidate the selector returns when its supervised term is averaged over $\mathcal{S}_{-k}$ only, the distortion term being unchanged because it uses no labels. The retained coefficient is

$$\hat{c} = \arg \min_{c \in \mathcal{C}} \frac{1}{F} \sum_{k=1}^F L^{\text{cv}}(\hat{g}_{c,-k}, \mathcal{S}_k)$$

with exact ties resolved by the first value in the ordered grid, and the reported candidate is the one the selector returns on all of $\mathcal{S}$ at $\hat{c}$. Algorithm 1 states the whole procedure.

The held-out loss L^{cv} is the selector’s own supervised criterion: empirical cross-entropy for CE-combo, and cross-entropy for alignment and for the permutation control, whose directional statistic is a score component rather

than a loss. Under corruption the labels in both $\mathcal{S}_{-k}$ and $\mathcal{S}_k$ are the corrupted ones, so no selector sees a clean label at any stage. The anchored-term grid is $\alpha \in \{0, 0.05, 0.1, 0.25, 0.5, 1, 2, 4, 8, 16, 32, 64\}$ (12 values) and the alignment grid is $\lambda \in \{0, 0.5, \dots, 10\}$ (21 values). Both contain zero, so an anchored selector can revert to pure distortion; because α multiplies the supervised term, large α approaches direct validation in the limit. On the finite grid the largest value is selected in 3.3%, 2.7% and 10.0% of outcomes at ten, fifty and all labels, and the selector at $\alpha_{\max} = 64$ returns the direct-validation candidate in 98.0%, 95.3% and 100% of them, so the grid spans the interpolation to that resolution. Every mechanism variant selects its own coefficient through the identical procedure, so differences between them are not attributable to tuning freedom.

Supplement Table S4 lists the four settings and their sizes.

How the medians of the main-text tables are taken. The medians reported for the three selectors across budgets are computed after averaging the repeated labeled subsets within each family-run: they are medians of run means rather than medians of the pooled run-subset outcomes. The two differ because the median does not commute with the mean, and the pooled median is lower at the smallest budgets. The run-level convention is the one stated in Supplement B and the one the paired inference uses.

When the anchor is already optimal. The label-free pick is not uniformly suboptimal. Read run by run, the candidate that minimizes teacher distortion attains zero cross-entropy regret in 15 of the 134 families, ten of them on MobileNetV2, three on ResNet50 and two on ViT-DomainNet, and is non-oracle in the remaining 119. The aggregate regrets reported elsewhere are means over those two groups, so the label-free rule is best read as usually, but not always, leaving regret on the table.

Inferential unit. Figure 2 summarizes the design. Unless a caption explicitly identifies a pooled run-subset or run-realization distribution, descriptive means and medians are computed after averaging repeated outcomes within each family-run. Tables reporting upper tails or operational outcome distributions pool the repetitions, as their captions state. Formal inference never treats subsets, candidates or corruption realizations as independent units. Descriptive quantities, all means and medians reported for the CNN settings, are computed over the 45 teacher-family runs, that is five shifts by three seeds by three executions, and over 17 runs on ViT-DomainNet and 27 on ViT-CIFAR-20. Paired inference is computed after averaging the three executions within each shift-seed configuration, which leaves 15 paired units per CNN architecture, because the three executions share splits and are not independent replicates of the same comparison. Neither level counts candidates or resample outcomes as units.

Where the between-run variation comes from. The standard deviations reported in Table 1 are large relative to their means, and a variance decomposition shows why. Taking CE-combo at ten labels and grouping the 45 runs by target shift, the between-shift component accounts for 90% of the

total variance on ResNet50 and 74% on MobileNetV2. The dispersion is therefore mostly heterogeneity across domain shifts, a property of the benchmark, rather than instability of the selector on a given shift.

Clustering. The 15 base configurations of each CNN setting are five shifts crossed with three seeds, and three of the five shifts share Sketch as target domain, so the runs are not fully exchangeable. Run-level bootstrap intervals should be read as having approximately five effective clusters per architecture, and the exact sign-flip test on 15 units has limited power. We therefore treat per-setting non-significance as inconclusive rather than as evidence of no effect, and treat the per-setting values as the primary evidence. Supplement Table S5 reports the resulting paired comparisons between supervised selectors.

What each pipeline stores. Every pipeline stores predictive distributions for the teacher and each of its candidates on all three splits. Every estimator we screen is defined on those distributions and is evaluated on all four settings.

Computing infrastructure. Experiments were run in Google Colab notebooks on Python 3.12.13 under Ubuntu 22.04.5 LTS, kernel 6.6.122, with results written to Google Drive. Colab allocates the host per session, so the runtime is a class rather than a fixed machine: the sessions used 2 to 8 vCPUs of an Intel Xeon at 2.0–2.2 GHz with 12 to 50 GB of system memory, and an NVIDIA accelerator, observed as A100-SXM4 with 80 GB and Tesla T4 with 15 GB, driver 580.82.07 and CUDA toolkit 12.8. The two DomainNet CNN pipelines use TensorFlow 2.20.0, with teachers initialized from ImageNet weights supplied by the framework’s application module; the two ViT pipelines use PyTorch 2.11.0 with torchvision 0.26.0 and timm 1.0.28. Quantization is simulated in floating point and applies to weights only, so no integer kernel or quantization-specific runtime is required. The selection and analysis stage consumes only the stored probability matrices and runs on CPU with NumPy 2.0.2 and SciPy 1.16.3: the principal selector tables recompute from the archived outcome files and individual runs from the stored predictions, both without an accelerator, while the full-cohort diagnostics and the deployment-cost analysis need the complete cache collection or architecture-level metadata.

C Calibration and Candidate-Family Sensitivity

On ResNet50, validation CE selects candidates with substantially lower ECE and cross-entropy but lower top-1 accuracy; on MobileNetV2 it reduces both while producing a small positive accuracy difference. This explains why validation accuracy fails as a corruption-robust criterion on ResNet50 specifically: the quantity that must be recovered there is invisible to a 0–1 loss.

Distortion remains poor on ResNet50 after restricting the family to high-precision candidates or to the top 20 by distortion, and its median regret rises to 0.471 inside the low-precision family, so the selection difficulty is not an artifact of a single catastrophic low-bit candidate. The comparison

Table S3: Selectors and ablations. All supervised entries receive the same labeled sample $\mathcal{S}$ of the stated size, including labels consumed by coefficient selection.

Selector	Score minimized over $g \in \mathcal{G}$	Role
Distortion	$D_u^{\text{KL}}(g, f)$	label-free teacher-relative baseline
Highest precision	none (fixed configuration)	data-free baseline matching the distortion pick
Validation CE	$\widehat{\text{CE}}_{\mathcal{S}}(g)$	direct supervised reference
CE-combo	$D_u^{\text{KL}}(g, f) + \alpha \widehat{\text{CE}}_{\mathcal{S}}(g)$	anchored cross-entropy
Alignment	$D_u^{\text{KL}}(g, f) - \lambda A_{\mathcal{S}}^{\text{B}}(g, f)$	teacher-relative directional selector
Teacher component	$D_u^{\text{KL}}(g, f) - \lambda A_{\mathcal{S}}^{\text{teach}}(g, f)$	mechanism ablation, no observed class
Permutation control	$D_u^{\text{KL}}(g, f) - \lambda A_{\mathcal{S}}^{\pi}(g, f)$	destroys residual-direction–input correspondence

Table S4: Experimental settings. A run contains one fixed teacher and its 72-candidate family. The selection pool is the union of the quantization-calibration and selection splits; the largest label budget is listed separately because it is not the pool size. Exec. counts independent end-to-end repetitions of the whole pipeline with retrained teachers on identical splits.

Data	Teacher	Cls.	Cfg.	Ex.	Runs	Pool	Max. n
DomainNet	ResNet50	30	15	3	45	300	300
DomainNet	MNV2	30	15	3	45	300	300
DomainNet	ViT	50	17	1	17	2.5k–6.3k	150
CIFAR-20	ViT	20	27	1	27	1332	200
Total	–	–	74	–	134	–	–

does not show uniform dominance of any anchored method across settings and budgets.

Anchor placement and anchor choice

Two design questions can be answered by re-running the same protocol with one element changed, and Supplement Table S7 does so. Evaluating the teacher distortion on the complete unlabeled pool, on the labeled subsample, or on an independent subsample of equal size gives regrets that differ by at most 0.007, well inside the run-to-run standard deviation, so nothing in our conclusions turns on that choice even though the operational argument favours using all available inputs. Replacing the teacher anchor with an optimal-transport anchor is more informative: it improves on the teacher on MobileNetV2, from 0.107 to 0.085 at ten labels, and worsens it on ResNet50, from 0.163 to 0.189. Anchoring is therefore a general construction rather than a property of the teacher, and each anchor helps where it is itself the stronger label-free rule; the teacher’s attraction is that it is immune to the confidence-collapse mode that defeats the confidence-based family.

Clipping and metric sensitivity

Supplement Table S8 reports two checks. Varying the probability floor across three values spanning six orders of magnitude moves distortion’s regret by at most 0.0003, so the large regret magnitudes are a property of the candidate families and not of the floor. Supplement Table S6 evaluates the label-free screen on accuracy regret rather than cross-entropy regret, addressing the objection that several estimators were

Table S5: Paired run-level inference. Differences are computed after averaging within-run outcomes; negative favors the first method. p is the two-sided sign-flip value, obtained by complete enumeration of the sign assignments when the number of paired units is at most 22 and by a 200,000-draw Monte Carlo sign-flip test otherwise, Holm-adjusted across the comparisons shown in each row, which is the family; boldface marks $p < 0.05$ after adjustment. Upper block: clean labels. Lower block: symmetric corruption.

Setting	n	CE-cmb – Val-CE		Align – Val-CE		Align – CE-cmb	
		Diff.	p	Diff.	p	Diff.	p
RN50–DN	10	-0.016	1.000	-0.014	1.000	+0.001	1.000
	25	+0.003	0.776	+0.020	0.469	+0.016	0.077
	50	+0.008	0.393	+0.026	0.112	+0.018	0.063
	100	+0.004	0.485	+0.020	0.167	+0.016	0.029
	150	-0.001	0.899	+0.012	0.324	+0.012	0.324
	300	-0.006	0.879	-0.001	0.879	+0.005	0.879
MNV2–DN	10	-0.124	<0.001	-0.139	<0.001	-0.015	0.111
	25	-0.039	<0.001	-0.037	<0.001	+0.002	0.448
	50	-0.012	0.158	-0.009	0.188	+0.004	0.215
	100	-0.001	1.000	-0.000	1.000	+0.001	1.000
	150	+0.001	1.000	+0.000	1.000	-0.001	1.000
	300	+0.004	1.000	+0.003	1.000	-0.000	0.938
Setting	η	Diff.	p	Diff.	p		
RN50–DN	20%	-0.010	<0.001	-0.011	0.102		
	40%	-0.032	<0.001	-0.051	<0.001		
MNV2–DN	20%	-0.076	0.013	-0.166	0.013		
	40%	-0.061	0.012	-0.431	0.002		
ViT–DN	20%	-0.043	0.248	-0.021	0.768		
	40%	-0.016	0.123	-0.295	0.036		
ViT–C20	20%	-0.069	0.006	-0.000	0.913		
	40%	+0.000	1.000	+0.000	1.000		

designed to rank accuracy: the anchor incurs 0.003 and 0.008 in accuracy regret while average confidence incurs 0.318 and 0.310, so the failure is not an artifact of scoring them on an objective they were not built for.

Supplement Table S8 collects both checks.

D Additional Experimental Results

Budget sweep and upper tail. Supplement Table S9 gives every clean-label budget we evaluate. Reading down a setting shows the pattern the main text summarizes: anchoring

Table S6: Accuracy regret, the quantity several of the label-free estimators were proposed to rank, over all three executions and 45 runs per architecture. Entries are the gap in target top-1 accuracy between the best candidate and the one selected. The deployment objective in this paper is cross-entropy, so these estimators are elsewhere evaluated outside their original objective; on their own metric the confidence-based group still fails by two orders of magnitude. The anchored selectors spend labels, so they are outside this label-free comparison.

Estimator	RN50-DN	MNV2-DN
Distortion	0.0032	0.0081
SoftmaxCorr	0.0092	0.0085
Nuclear norm	0.0090	0.0031
DOC	0.0033	0.0076
ATC-MC	0.3211	0.2717
ATC-NE	0.3211	0.2404
Entropy	0.3178	0.3101
Average confidence	0.3178	0.3101

Table S7: Where the anchor is evaluated, and which anchor. Mean regret over the first complete execution of each CNN setting, 15 runs per architecture, so the values are not directly comparable with the pooled means of Supplement Table S9. Columns two to four vary only the input set on which the teacher distortion is averaged: the complete unlabeled pool, the labeled subsample itself, or an independent subsample of the same size. The last column replaces the teacher anchor with the optimal-transport estimator, rescaled so that the coefficient grid remains comparable.

Setting	n	Val-CE	All $\mathcal{U}$	Same $\mathcal{S}$	Indep.	COT anchor
ResNet50-DomainNet	10	0.1801	0.1627	0.1564	0.1629	0.1888
	50	0.0608	0.0677	0.0682	0.0672	0.0757
	300	0.0129	0.0086	0.0086	0.0086	0.0094
MobileNetV2-DomainNet	10	0.2473	0.1065	0.1080	0.1101	0.0847
	50	0.0504	0.0231	0.0227	0.0219	0.0119
	300	0.0010	0.0010	0.0010	0.0010	0.0009

Table S8: Two sensitivity checks. Left: distortion’s cross-entropy regret under three probability floors, showing that the magnitudes do not depend on clipping. Right: regret in *accuracy*, the quantity the confidence estimators were designed to rank, for the anchor and for average confidence. The confidence estimator fails on its own metric as well.

Setting	CE regret at floor			Accuracy regret	
	10^{-6}	10^{-8}	10^{-12}	Dist.	Avg. conf.
ResNet50-DomainNet	0.2158	0.2161	0.2161	0.0040	0.3212
MobileNetV2-DomainNet	0.0848	0.0848	0.0848	0.0076	0.3076

wins at ten and twenty-five labels and loses from fifty onward, which is where the eleven-of-twenty-three count in the main text comes from. Supplement Table S10 gives the corresponding upper tail. The 95th percentile and the catastrophic-

Table S9: Budget ablation: mean target cross-entropy regret at every clean-label budget evaluated. Distortion is label-free and constant in n . Boldface marks the lowest of the three supervised selectors in each row. Anchoring wins at the smallest budget in all four settings and in seven of the remaining nineteen comparisons; its advantage concentrates at ten and twenty-five labels and fades from fifty onward. Runs are 45 per CNN architecture, pooling three executions, 17 on ViT-DomainNet and 27 on ViT-CIFAR-20.

Setting	n	Dist.	Val-CE	CE-combo	Align
ResNet50-DomainNet	10	0.2198	0.1819	0.1664	0.1678
	25	0.2198	0.1015	0.1047	0.1211
	50	0.2198	0.0619	0.0702	0.0882
	100	0.2198	0.0334	0.0372	0.0532
	150	0.2198	0.0282	0.0276	0.0398
	300	0.2198	0.0132	0.0071	0.0121
MobileNetV2-DomainNet	10	0.0873	0.2270	0.1026	0.0875
	25	0.0873	0.0828	0.0440	0.0462
	50	0.0873	0.0350	0.0227	0.0265
	100	0.0873	0.0130	0.0116	0.0129
	150	0.0873	0.0075	0.0089	0.0078
	300	0.0873	0.0012	0.0049	0.0045
ViT-DomainNet	10	0.2843	0.2078	0.1395	0.2046
	25	0.2843	0.0912	0.0822	0.1273
	50	0.2843	0.0593	0.0634	0.0860
	100	0.2843	0.0287	0.0343	0.0709
	150	0.2843	0.0165	0.0215	0.0511
ViT-CIFAR-20	10	0.0955	0.2215	0.1155	0.1496
	25	0.0955	0.0998	0.0880	0.1139
	50	0.0955	0.0541	0.0636	0.0821
	100	0.0955	0.0373	0.0550	0.0693
	150	0.0955	0.0270	0.0458	0.0599
	200	0.0955	0.0260	0.0443	0.0517

selection rate move together with the mean at the smallest budgets and separate at the largest, where anchoring can leave the mean unchanged while still removing the worst outcomes.

Paired inference and coefficient stability. Supplement Table S5 reports the paired run-level tests. Under clean supervision the selected coefficient is generally away from the upper boundary; under 40% corruption on MobileNetV2 the grid binds frequently, reaching $\lambda_{\max}$ in 53% of runs; Supplement Table S11 gives the distribution of the coefficient and Supplement Figure 3 the regret curve behind it. On ViT-CIFAR-20 at $\eta = 0.4$ the paired differences are exactly zero because the three supervised selectors select the same candidate in every run at that corruption level, so the comparison carries no information there.

Corruption at the complete pool. Supplement Table S12 isolates corruption from scarcity by fixing the budget at the complete labeled pool. It evaluates the same $n = 300$ conditions as the budget-rate grid but pools all three executions, so its cohort is larger. It reports medians alongside means and shows most clearly that at $\eta = 0.4$ the directional variant attains the lowest median on both architectures.

Table S10: Upper-tail behavior across clean-label budgets, pooling all three end-to-end executions, 45 runs per architecture. Left block: 95th percentile of target cross-entropy regret pooled over run-subset outcomes rather than over run means, since the tail is a property of the outcome distribution. Right block: the catastrophic-selection rate $\Pr(R > 0.1)$. The last column is the complete labeled pool of 300.

Setting	Selector	95th-percentile regret						$\Pr(R > 0.1)$					
		10	25	50	100	150	300	10	25	50	100	150	300
RN50-DN	Val-CE	0.58	0.35	0.24	0.14	0.12	0.08	0.50	0.33	0.17	0.07	0.06	0.02
	CE-combo	0.49	0.37	0.33	0.20	0.13	0.03	0.49	0.32	0.19	0.08	0.06	0.00
	Align	0.48	0.41	0.34	0.28	0.18	0.05	0.50	0.36	0.27	0.13	0.09	0.00
MNV2-DN	Val-CE	1.18	0.61	0.13	0.06	0.04	0.00	0.30	0.15	0.06	0.02	0.01	0.00
	CE-combo	0.71	0.20	0.12	0.08	0.06	0.02	0.23	0.13	0.06	0.02	0.02	0.00
	Align	0.44	0.20	0.13	0.09	0.06	0.01	0.25	0.14	0.08	0.03	0.01	0.00

Table S11: Distribution of the cross-validated alignment coefficient at the complete labeled pool. $\Pr(\lambda=0)$ is the frequency with which the procedure declines to use the supervised direction and reverts to pure distortion; $\Pr(\lambda=\lambda_{\max})$ the frequency of selecting the grid boundary. SD is across run medians.

Setting	Supervision	Median λ	IQR	$\Pr(\lambda=0)$	$\Pr(\lambda=\lambda_{\max})$	SD
ResNet50-DomainNet	Clean	7.5	[5.5, 9.0]	0.111	0.156	3.12
	20% noise	9.0	[8.5, 9.5]	0.000	0.244	1.28
	40% noise	8.5	[7.5, 9.0]	0.000	0.067	1.01
MobileNetV2-DomainNet	Clean	3.5	[0.0, 5.5]	0.289	0.022	2.81
	20% noise	4.0	[0.0, 5.5]	0.289	0.000	2.91
	40% noise	10.0	[8.0, 10.0]	0.133	0.533	3.41
ViT-DomainNet	Clean	5.5	[2.0, 6.0]	0.059	0.000	2.63
	20% noise	7.5	[5.5, 8.0]	0.000	0.059	3.23
	40% noise	7.0	[6.0, 8.0]	0.000	0.059	1.32
ViT-CIFAR-20	Clean	6.5	[5.8, 7.5]	0.000	0.000	1.60
	20% noise	6.5	[5.0, 8.0]	0.000	0.000	1.50
	40% noise	7.5	[6.8, 7.5]	0.000	0.000	0.64

Table S12: Symmetric label corruption at the complete labeled pool, pooling all three end-to-end executions, 45 runs per architecture: median target cross-entropy regret with mean in parentheses taken over runs after averaging corruption realizations within each run. Boldface marks the lowest median in each row.

Setting	η	Val-CE	CE-combo	Align
RN50-DN	20%	0.064 (0.078)	0.064 (0.068)	0.065 (0.067)
	40%	0.165 (0.231)	0.127 (0.200)	0.102 (0.180)
MNV2-DN	20%	0.009 (0.181)	0.013 (0.105)	0.001 (0.015)
	40%	0.779 (0.831)	0.706 (0.770)	0.393 (0.399)

The directional term and its control. Supplement Table S13 reports the permutation control and the two components of the alignment statistic. The control’s column is the one to read against distortion: it tracks the anchor to within 0.025 everywhere, which is what makes the comparison informative only where reverting to the anchor is costly. The teacher component is the best of the three variants on ResNet50 at ten labels and the worst elsewhere at that budget.

Convention of the permutation control. The implemented control, stated in Table S1 and in the caption of Table S13, permutes the entire teacher-residual direction $e_y - p_f(x)$ across the labeled examples, so the displacement at x_i is scored against the complete target direction of another example; the main text abbreviates this as permuting the observed labels. The two constructions differ: permuting labels alone would leave the teacher component $\langle \delta_g^p(x_i), p_f(x_i) \rangle$ tied to its input, and Table S13 shows that this component carries selection signal of its own, so a labels-only permutation is not a null for the full directional statistic. The residual-direction permutation is: it destroys the correspondence between every input and its own target direction while preserving the marginal distribution and scale of the statistic, and its behaviour in Table S13, tracking pure distortion, is what a proper null predicts. The question a labels-only permutation would address, namely what the observed-label part contributes relative to the teacher part, is answered directly by the $A_S^{\text{lab}}/A_S^{\text{teach}}$ decomposition reported in the same table. All permutation results in this paper use the residual-direction construction.

Deployment cost. Supplement Table S14 gives the price of each memory budget and the spread that survives inside it.

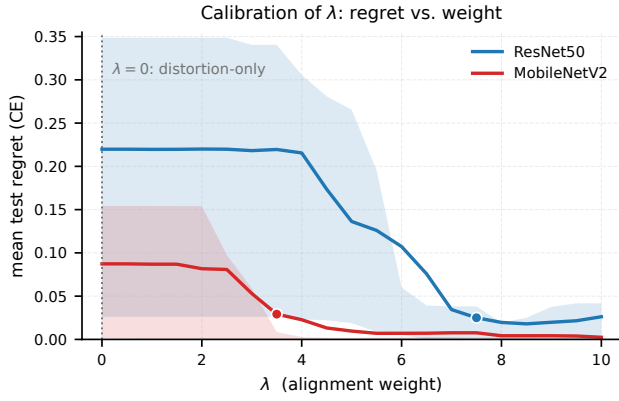


Figure 3: Mean target cross-entropy regret as a function of the alignment weight λ on the two DomainNet CNN settings. $\lambda = 0$ recovers distortion-only selection. Mean regret decreases with λ on both settings, steeply on ResNet50 between roughly $\lambda = 4$ and $\lambda = 8$ and more gradually thereafter; the shaded bands are interquartile ranges across runs. Marked points are the medians of the cross-validated coefficient.

Table S13: The permutation control and the two components of the directional term, in median target cross-entropy regret. The control receives the identical score structure, coefficient grid and cross-validation as alignment, and differs only in that the teacher-residual directions $e_y - p_f(x)$ are permuted across the labeled examples before the statistic is formed. Boldface marks the lowest of the three in each row. CNN rows pool all three executions, 45 runs per architecture.

Setting	n	Dist.	Align	Perm.	Teach.
ResNet50–DomainNet	10	0.2323	0.1870	0.2170	0.1445
	25	0.2323	0.1143	0.2228	0.1101
	50	0.2323	0.0635	0.2276	0.0692
	100	0.2323	0.0342	0.2304	0.0432
	150	0.2323	0.0270	0.2320	0.0360
	300	0.2323	0.0024	0.2307	0.0260
MobileNetV2–DomainNet	10	0.0778	0.0816	0.0584	0.2169
	25	0.0778	0.0399	0.0703	0.1458
	50	0.0778	0.0241	0.0685	0.0754
	100	0.0778	0.0049	0.0653	0.0568
	150	0.0778	0.0026	0.0662	0.0314
	300	0.0778	0.0000	0.0653	0.0021
ViT–DomainNet	10	0.0200	0.0501	0.0231	0.1957
	25	0.0200	0.0434	0.0213	0.0796
	50	0.0200	0.0295	0.0175	0.0387
	100	0.0200	0.0183	0.0181	0.0176
	150	0.0200	0.0110	0.0173	0.0091
ViT–CIFAR-20	10	0.1009	0.1325	0.0912	0.2162
	25	0.1009	0.1010	0.1056	0.1270
	50	0.1009	0.0764	0.0966	0.0745
	100	0.1009	0.0715	0.0988	0.0538
	150	0.1009	0.0531	0.1002	0.0457
	200	0.1009	0.0458	0.1006	0.0398

Table S14: Selection under a deployment-cost constraint. Cost is the weight memory of a candidate relative to full precision, with layers kept in full precision counted at their true width. For each budget we report how many of the 72 candidates are admissible, its excess over the unconstrained best, and the mean spread between the worst and best admissible candidate. Means over 45 runs per architecture.

Setting	Budget	Admissible	Excess over free	Internal spread
RN50–DN	$\leq 10\%$	12	0.710	14.41
	$\leq 16\%$	36	0.103	15.02
	$\leq 22\%$	54	0.007	15.12
	none	72	0.000	15.13
MNV2–DN	$\leq 10\%$	12	0.901	10.41
	$\leq 16\%$	36	0.869	10.63
	$\leq 22\%$	54	0.868	10.63
	none	72	0.000	11.50

The two columns answer different questions: the excess over the unconstrained best is what the constraint costs, while the internal spread is what selection can still recover once the constraint is imposed, and only the first varies much across settings. Supplement Table S15 then restricts the candidate set, the candidate search, the coefficient search and the oracle to the admissible family; the labeled sample is unchanged, so its regrets are measured against the best admissible candidate rather than the best overall.

E Degradation Regimes and the Failure of Confidence-Based Estimators

This section of the supplement supports two claims made in Section 4 of the main paper. The confidences quoted there for ResNet50, 1.000 for the worst candidate against 0.408 for the best, are the medians reported in Table S16; and the statement that the two ViT settings invert the pattern is quantified here over all executions of the four settings rather than the single run shown in Figure 1.

Section 4 attributes the failure of the confidence-based estimators to the degradation regime rather than to the estimators themselves. This section gives the evidence for that attribution, quantifies how far threshold calibration can compensate, and states why we stop short of a mechanism. All quantities are computed from the stored predictive distributions; no retraining or requantization is involved.

The two regimes, quantified. Table S16 reports for each setting, over all executions, the median regret of the four confidence-based estimators alongside the diagnostics that account for it. The inversion described in the body is visible in three quantities at once. Accuracy and mean confidence across the 72 candidates are uncorrelated or inversely related on the CNNs and positively related on the ViTs; the worst candidate is more confident than the best on the CNNs and less confident on the ViTs; and the confidence-based estimators select a chance-accuracy candidate in most runs of each CNN architecture and in none on the ViTs. The rate is not uniform within the group: on MobileNetV2 the two

Table S15: Constrained selection. The candidate set, the coefficient search and the oracle are all restricted to $\mathcal{G}_B = \{g : C(g) \leq B\}$; the labeled sample is unchanged, so regret is measured against the best admissible candidate. Pooled over all three executions, 45 runs per architecture, lower of the two supervised selectors in boldface.

Setting	Budget	$n=10$			$n=50$			$n=300$		
		Dist.	Val-CE	CE-combo	Dist.	Val-CE	CE-combo	Dist.	Val-CE	CE-combo
ResNet50–DomainNet	$\leq 10\%$	0.095	0.097	0.090	0.079	0.043	0.042	0.075	0.002	0.004
	$\leq 16\%$	0.305	0.168	0.160	0.304	0.040	0.056	0.304	0.014	0.016
	$\leq 22\%$	0.269	0.184	0.176	0.283	0.063	0.078	0.290	0.015	0.009
	none	0.220	0.182	0.164	0.220	0.062	0.070	0.220	0.013	0.007
MobileNetV2–DomainNet	$\leq 10\%$	0.011	0.003	0.005	0.000	0.000	0.000	0.000	0.000	0.000
	$\leq 16\%$	0.318	0.229	0.193	0.167	0.012	0.011	0.163	0.000	0.000
	$\leq 22\%$	0.437	0.310	0.308	0.348	0.056	0.096	0.331	0.003	0.007
	none	0.076	0.227	0.102	0.086	0.035	0.022	0.087	0.001	0.005

ATC variants do so in 39 and 35 of 45 runs while average confidence and entropy do so in all 45.

Threshold calibration mitigates without repairing. The two CNN settings differ in how far calibration helps, and the difference tracks confidence saturation. On ResNet50 the collapsed candidates take only 34% distinct confidence values and reach exactly 1.000; the ATC threshold therefore falls inside a mass of ties, its accuracy estimate loses discriminative power, and the median regret of all four estimators is approximately the median family spread. On MobileNetV2 confidences remain distinct, calibration roughly halves the median regret ($10.81 \rightarrow 4.89$), and no estimator selects the single worst candidate. ATC-MC nevertheless selects a chance-accuracy candidate in 39 of 45 runs and ATC-NE in 35 of 45. Threshold calibration addresses the symptom, not the cause.

Where collapse lives in the candidate grid. Table S17 breaks the collapse rate down by bit width and scaling granularity. One regularity is common to all four settings: collapse concentrates at two and three bits. Granularity matters only at intermediate widths and not uniformly: per-tensor collapses more often than per-channel at three and four bits on ResNet50 and at five bits on MobileNetV2, while at two bits the two are comparable and on ViT–CIFAR-20 the order reverses. MobileNetV2 is the exception in degree rather than in kind: it still collapses at four and five bits, so a substantial part of its nominal 72-candidate family consists of degenerate predictors. This does not affect reported regrets, which are measured against the best candidate actually attained, but it qualifies what “72 candidates” means in that setting.

We do not claim a mechanism. The natural explanation for the split in Table S16 is normalization: the CNNs carry frozen BatchNorm statistics that become stale when weights are quantized, whereas LayerNorm normalizes at inference and cannot inherit stale statistics. Our design cannot support that claim, for two reasons. First, architecture and teacher provenance are confounded: every BatchNorm model here is obtained by transfer learning against hard labels, and every LayerNorm model comes from a temperature-based distillation pipeline (distillation temperatures 2.0 and 10.0) whose outputs are consistent with the different soft-target training

regime, although the objective imposes no hard confidence bound. The maximum confidence observed among collapsed two-bit candidates is 0.8008 on ViT–CIFAR-20 and 0.7998 on ViT–DomainNet, against 1.0000 on ResNet50 and 0.9998 on MobileNetV2, a ceiling consistent with soft-target training rather than with LayerNorm. Second, the BatchNorm account does not separate the two CNNs: ResNet50 shows a confidence gap of $+0.704$ between collapsed and healthy candidates, MobileNetV2 only $+0.069$, although both carry BatchNorm, and MobileNetV2 collapses more often while saturating less. Separating the two would require a model that breaks the pairing, a LayerNorm backbone trained against hard labels or a convolutional one distilled with soft targets, which is a different construction from the candidate families this paper selects over. We therefore report the phenomenon and its consequence, and leave the mechanism open.

Consequence for practice. The asymmetry matters more than the average. Confidence-based selection is adequate in one regime and catastrophic in the other, and deciding which regime holds requires the target labels the setting denies: the diagnostics in Table S16 that separate the regimes (accuracy of the worst candidate, the accuracy–confidence correlation) are themselves label-dependent. Teacher distortion is unaffected in either regime, because a collapsed candidate is far from the teacher’s output distribution however confident it is. This is the sense in which anchoring to the teacher is a safer default rather than merely a better average performer.

F Cohort Inclusion Criteria

A run enters the final cohort only if it yields a complete 72-candidate family with usable stored predictions. The criteria are structural and were applied to the runs available during the audit, before any comparative selector result from those pipelines could enter the tables. A run is excluded if its candidate family is incomplete, if its per-run outputs are uniform across candidates, or if every candidate sits at chance accuracy.

One run falls outside it. Of the 18 ViT–DomainNet runs generated, one contains 47 of the 72 candidates; the missing indices 47 to 71 form a contiguous block at the end of the enumeration order, so the exclusion cannot depend on

Table S16: Degradation regimes. Median target cross-entropy regret of the four confidence-based estimators, with diagnostics. ρ is the median Pearson correlation between test accuracy and mean confidence across the 72 candidates within a run. c_{worst} and c_{best} are the median confidences of the worst and best candidate by test cross-entropy. “Distinct” is the median fraction of distinct confidence values among collapsed candidates (accuracy $\leq 1.5\times$ chance). CNN rows pool three end-to-end executions.

Setting	Runs	K	AvgConf	Entropy	ATC-MC	ATC-NE	ρ	c_{worst}	c_{best}	Distinct
ResNet50–DomainNet	45	30	15.062	15.062	15.062	15.062	−0.38	1.000	0.408	34%
MobileNetV2–DomainNet	45	30	10.814	10.814	4.886	4.696	−0.09	0.985	0.534	100%
ViT–CIFAR-20	27	20	0.124	0.167	0.100	0.099	+0.86	0.579	0.726	100%
ViT–DomainNet	17	50	0.062	0.339	0.020	0.020	+0.88	0.523	0.761	100%

Table S17: Collapse rate, the fraction of candidates with accuracy at most $1.5\times$ chance, by bit width and scaling granularity. PT is per-tensor and PC per-channel. Percentages are over candidates and runs within each setting.

Bits	RN50		MNV2		ViT-C20		ViT-DN	
	PT	PC	PT	PC	PT	PC	PT	PC
2	99	100	100	100	86	96	91	88
3	36	16	99	100	33	0	31	0
4	33	0	91	90	15	0	17	0
5	0	0	87	39	0	0	0	0
6	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	0

measured performance. It does not enter any table.

The cohort reported throughout is therefore 45 runs per CNN architecture and 17 on ViT–DomainNet and 27 on ViT–CIFAR-20, as Table S2 states: the full design of Section 4, and the evidence behind every claim we report.